PRE-ALGEBRA (7TH) · DATA ANALYSIS AND PROBABILITY

Spinner Probability Lab

Explore & Reflect **TEACHER KEY**

Name _____ Date _____ Period _____

Open **Spinner Probability Lab** in your unit's Interactive Math section on Mr. K's Classroom (or use your teacher's link). Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3. Do Parts 4 and 5 on your own.

1 Predict before you touch the tool

1. A spinner has 3 red parts, 2 blue parts and 1 green part, all the same size. What is P(red)?
   $\frac{3}{6} = \frac{1}{2}$
2. If you spin 10 times, will you get exactly 5 reds? If you spin 1,000 times, about how many reds do you expect?
   Not necessarily. About 500.

2 Explore use the tool

1. Clear the results and spin 10 times. Do this three times. Record the share of red each time.
   Answers vary, e.g., 0.3, 0.6, 0.5. Small samples bounce around 0.5.
2. Spin 1,000 or more times. How close is each color to its theoretical probability?
   Answers vary; usually within about 0.03.
3. Build a spinner with P(red) = $\frac{1}{4}$ that uses 3 colors. List your parts.
   Answers vary, e.g., red 1, blue 2, green 1 (or red 2, blue 3, green 3).
4. Red came up 4 times in a row. Is blue more likely on the next spin? Explain.
   No. The spinner has no memory; P(blue) is the same on every spin.

3 Record use the tool and fill in the table

Theoretical and expected results

Parts are listed as Red, Blue, Green, Yellow.

Parts (R, B, G, Y)	P(red)	P(not red)	Expected reds in 300 spins
3, 2, 1, 0	$\frac{1}{2}$	$\frac{1}{2}$	150
1, 1, 1, 1	$\frac{1}{4}$	$\frac{3}{4}$	75
2, 5, 1, 0	$\frac{1}{4}$	$\frac{3}{4}$	75
4, 1, 0, 1	$\frac{2}{3}$	$\frac{1}{3}$	200
0, 3, 3, 2	0	1	0

4 Explain what did you discover?

1. Theoretical probability = parts for that color ÷ total parts.
2. Experimental probability = times it happened ÷ total spins.

3. As the number of spins grows, the experimental probability usually gets **closer** to the theoretical probability.

4. Expected number of results = probability × **number of spins**.

5. Maya spun 20 times and got red 15 times, so she says P(red) = $\frac{3}{4}$ for a spinner with 3 red parts out of 6. Is she right? Explain.
   No. $\frac{15}{20}$ is her experimental probability. The theoretical probability comes from the parts: $\frac{3}{6} = \frac{1}{2}$. With only 20 spins, results can be far from it.

5 Apply a new situation, no tool

1. A bag has 4 red, 6 blue and 10 green marbles. Find P(blue). If you draw (and replace) 50 times, about how many blues do you expect?
   $\frac{6}{20} = \frac{3}{10}$; about 15 blues.

2. A number cube is rolled 600 times. About how many 5s do you expect?
   About 100.

3. A spinner with 4 equal parts (1 green) landed on green 18 times in 60 spins. Find the experimental P(green). Is the result reasonable?
   $\frac{18}{60}$ = 0.3. Theoretical is 0.25, so yes: close for 60 spins.

Teacher notes

Intended discovery: Theoretical probability comes from the spinner (parts ÷ total); experimental shares vary a lot for small samples and settle near the theoretical value as the number of spins grows. Each spin is independent.

Likely misconceptions:

- Expecting exact results in small samples (exactly 5 reds in 10).
- Gambler's fallacy: a color is "due" after a streak.
- Treating experimental results as the probability of the spinner.

Answers that can vary: All simulation results (Explore 1–2) and spinner designs (Explore 3).